

PROBABILITY THEORY AND RANDOM PROCESSES(MA225)

LECTURE SLIDES

Lecture 01

Books

- Text Book
 - *Introduction to Probability Models* by Seldon M. Ross.
- Reference Books
 - *Probability and Random Processes* by G. R. Grimmett and D. R. Stirzaker.
 - *One Thousand Exercises in Probability* by G. R. Grimmett and D. R. Stirzaker.
 - *An Introduction to Probability Theory and its Applications* by W. Feller.

Grading Policy

- Weights in different examination are as follows:
 - Quiz I(24/8/24, Objective Type): 10%
 - Quiz II(7/9/24, Objective Type): 10%
 - Mid-semester Examination(As per academic calender, Descriptive Type): 25%
 - Quiz III(19/10/24, Objective Type): 10%
 - Quiz IV(9/11/24, Objective Type): 10%
 - End-semester Examination(As per academic calender, Objective and Descriptive Type): 35%

Classical Probability

- S : Set of all possible outcomes.
- **Def:** $P(A) = \frac{\text{Favourable number of cases to } A}{\text{Total number of cases}} = \frac{\#A}{\#S}$.
- **Example 1:** A die is rolled. What is the probability of getting 3 on upper face?
▶ Ans: $1/6$.
- **Example 2:** Consider a target comprising of three concentric circles of radii $1/3$, 1 , and $\sqrt{3}$ feet. What is the probability that a shooter hits inside the inner circle?
▶ Both $\#A$ as well as $\#S$ are infinite, the classical probability can not be used here.

Remarks

- The classical definition works in the first example but does not work in the second.
- Need a better definition which works for wider class of models.
- Start with classical definition and take three key properties to give more general definition of probability.
- Define the probability as a set function.
- Define the domain properly.

Countability and Uncountability

Def: We say that two sets A and B are equivalent if there exists a bijection from A to B . We denote it by $A \sim B$.

Def: For any set A we say:

- ① A is countable if $A \sim \mathbb{N}$
- ② A is atmost countable if either A is finite or A is countable.
- ③ A is uncountable if A is not atmost countable.

Example 3: $\mathbb{Z}$ is countable.

Remark: If a set is countable, then it can be written as sequence $\{x_n\}$ of distinct terms.

Summary of Results

Theorem: Every subset of an atmost countable set is again atmost countable.

Theorem: Let $\{E_n\}_{n \geq 1}$ be a sequence of atmost countable sets and put $S = \bigcup_{n=1}^{\infty} E_n$. Then S is again atmost countable.

Theorem: Let $A_1, A_2, \dots, A_n$ be atmost countable sets. Then $B = A_1 \times A_2 \times \dots \times A_n$ is also atmost countable.

Corollary: The set of rationals is countable.

Theorem: The set of all binary sequences is uncountable.

Corollary: $[0, 1]$ is uncountable.

Corollary: $\mathbb{R}$ is uncountable.

Corollary: Q^c is uncountable.

Corollary: Any interval is uncountable.

Random Experiment

Def: An experiment is called a random experiment if it satisfies the following three properties:

- ① All the out comes of the experiment is known in advance.
- ② The outcome of a particular performance of an experiment is not known in advance.
- ③ The experiment can be repeated under identical conditions.

Example 4: Toss a coin.

Example 5: Toss a coin until the first head appears.

Example 6: Measuring the height of a student.

Sample Space

Def: The collection of all possible outcomes of a random experiment is called the sample space of the random experiment. It will be denoted by $\mathcal{S}$.

Example 7: $\mathcal{S} = \{H, T\}$.

Example 8: $\mathcal{S} = \{H, TH, TTH, \dots\}$

Example 9: $\mathcal{S} = (0, \infty)$

σ -algebra

Def: A non-empty collection, $\mathcal{F}$, of subsets of $\mathcal{S}$ is called a σ -algebra (or σ -field) if

- ① $\mathcal{S} \in \mathcal{F}$
- ② $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$
- ③ $A_1, A_2, \dots \in \mathcal{F}$ implies $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

Example 10: $\mathcal{F}_1 = \{\phi, \mathcal{S}, \{H\}, \{T\}\}$, $\mathcal{F}_2 = \{\phi, \mathcal{S}\}$,
 $\mathcal{F}_3 = \{\phi, \mathcal{S}, \{H\}\}$

Example 11: $\mathcal{F} = \mathcal{P}(\mathcal{S})$

Example 12: $\mathcal{F} = \{\phi, \mathcal{S}, (4, 5), (4, 5)^c\}$